Du Han

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PROFESSIONAL EXPERIENCE

Sensequake | *Software Engineer Internship* – Montreal, QC **May. 2024 – Aug. 2024**

- Developed a regression-based residualization approach to estimate and subtract temperature-related variation from structural vibration data, helping identify potentially abnormal vibration patterns.
- Processed time-series sensor data using data cleaning, interpolation, and feature selection; evaluated regression models using cross-validation and R^2 to support reliable analysis.
- Built interactive data visualization tools using Python, Plotly, and JavaScript to help engineers explore and interpret structural monitoring data.
- Migrated resource-intensive PDF report generation from AWS EC2 to Lambda using Docker and Amazon ECR, resolving runtime compatibility issues and reducing the impact on website performance.

EDUCATION

McGill University – Montreal, QC **Graduated Spring 2025**
Master of Science, Computer Science **Cumulative GPA: 3.81 / 4.0**

University of Toronto – Toronto, ON **Graduated Spring 2023**
Honours Bachelor of Science, Statistics (Machine Learning and Data Mining Stream) **Cumulative GPA: 3.57 / 4.0**

TECHNICAL SKILLS

Programming Languages: Python, SQL, R, Java, C++, C, C#, JavaScript, TypeScript
Data Science & ML Libraries: pandas, NumPy, SciPy, scikit-learn, PyTorch, TensorFlow/Keras
Machine Learning & Statistics: Regression, Classification, Ensemble Learning, Reinforcement Learning, Clustering, PCA, Regularization, Feature Engineering, Cross-Validation, Hypothesis Testing, Model Evaluation
Data & Visualization Tools: Matplotlib, Plotly, Jupyter Notebook, MySQL, PostgreSQL, MongoDB, SQLite
Cloud & Development Tools: AWS (EC2, S3, Lambda, ECR, Glue, Athena, Redshift), Docker, Git, Flask, REST APIs

PERSONAL PROJECTS

Deep Reinforcement Learning for Super Mario Bros. – Python (PyTorch/OpenAI Gym) – *Personal Side Project – 2024*

- Developed and trained a reinforcement learning agent using Deep Q-Networks (DQN) and convolutional neural networks to learn gameplay policies from visual observations.
- Implemented image preprocessing, including grayscale conversion, resizing, and frame stacking, to improve state representation and capture temporal information.
- Trained and compared DQN, Double DQN, and Dueling DQN architectures, analyzing reward curves and training efficiency across experiments processing over 50M frames.
- Evaluated optimizer and loss function configurations, achieving approximately 90% improvement in mean episodic reward over the initial baseline configuration.

Machine Learning for Nylon Crystallization Prediction – *Python, pandas, scikit-learn | Side Research Collaboration*

- Developed regression models to predict nylon crystallization outcomes from experimental parameters, including material composition ratios.
- Trained and evaluated Ridge Regression, Random Forest, and Gradient Boosting models, comparing their predictive performance and selecting Random Forest based on evaluation results.
- Preprocessed experimental data and collaborated with a materials science researcher to understand experimental requirements and interpret model predictions.

CERTIFICATIONS

AWS Certified Solutions Architect – Associate
Google IT Support Professional Certificate

LANGUAGES

English, Mandarin